

Gen AI Exchange Hackathon

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Problem Statement : AI Legal Simplifier – Demystifying Legal Documents

Brief about the prototype:

Our prototype, **AI Legal Simplifier**, is a web-based application that helps people easily understand complex legal documents. Users can upload agreements, contracts, policies, or government notices, and the system will automatically:

- Extract the text,
- Generate a **plain-language summary** in simple words,
- Provide **clause-by-clause explanations** with examples,
- Highlight important **obligations, penalties, and risks**, and
- Allow users to **ask questions** directly about the document.
- - The solution is powered by **Generative AI (LLMs)** integrated into a MERN stack application. It ensures that legal documents become **accessible, multilingual (English, Hindi, Marathi, etc.)**, and easy to understand for **students, citizens, and small businesses**.
- This prototype empowers people to **make informed decisions, avoid frauds, and save time/costs**, while also promoting **legal awareness and inclusivity**.

Opportunity should be able to explain the following:

How different is it from any of the other existing solutions?

Most existing tools only provide **generic text summarization**, not legal-specific explanations.

Current solutions lack **risk assessment** or **clause-by-clause clarity**.

Very few platforms offer **multilingual legal simplification**, limiting accessibility for non-English speakers.

Our solution goes beyond summarization → it provides **context, examples, and risk levels** tailored for legal documents.

How will it be able to solve the problem?

Simplifies complex legal documents into **plain, understandable language**.

Highlights **important obligations, penalties, and deadlines** so users don't miss critical terms.

Provides **interactive Q&A** for document-specific queries.

Makes legal knowledge **accessible to students, small businesses, and citizens** without requiring prior legal expertise.

Reduces the dependency on costly legal consultations for basic understanding.

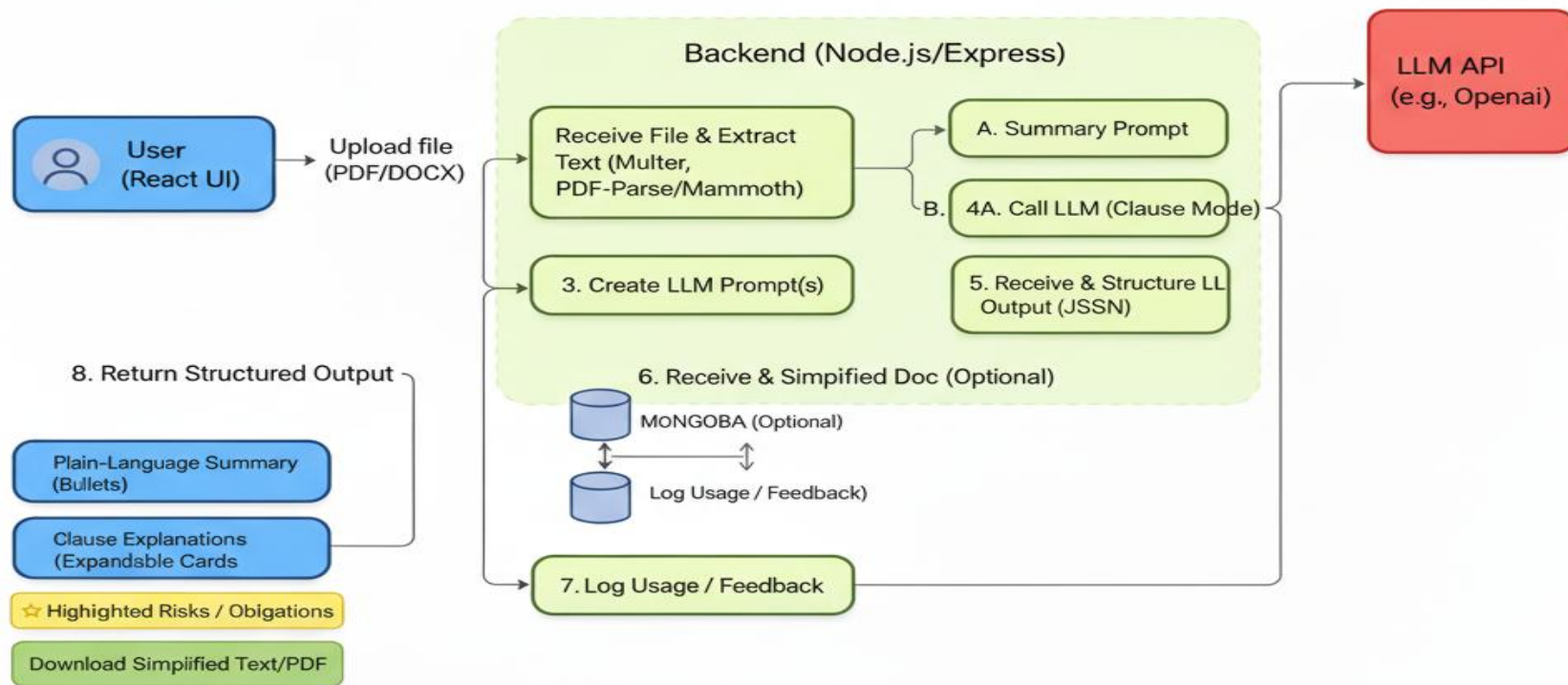
USP of the proposed solution

- ❑ **Legal-context AI simplification** (not generic summaries).
- ❑ **Risk categorization (Low/Medium/High)** for better decision-making.
- ❑ **Multilingual support** (English, Hindi, Marathi, etc.) for inclusivity.
- ❑ **Easy-to-use MERN web app** with privacy-first approach.
- ❑ **Scalable & adaptable** for contracts, policies, govt. schemes, and beyond.

List of features offered by the solution

- ❑ **Document Upload** – Supports PDF and DOCX formats for easy input.
- ❑ **Plain-Language Summary** – Converts complex legal text into simple, concise points.
- ❑ **Clause-by-Clause Explanations** – Breaks down each section with simplified meaning and practical examples.
- ❑ **Risk & Obligation Highlights** – Identifies critical terms such as penalties, deadlines, and liabilities.
- ❑ **Interactive Q&A Assistant** – Allows users to ask questions about specific parts of the document.
- ❑ **Multilingual Support** – Provides explanations in English, Hindi, Marathi (extensible to other languages).
- ❑ **Download & Export** – Users can save simplified versions in text/PDF format.
- ❑ **Privacy First** – No documents stored without user consent; local processing/logging options.
- ❑ **User Feedback Integration** – Feedback loop to improve explanations and accuracy over time.
- ♣ **Scalable Cloud-Based Deployment** – Accessible via web browser, no installation needed.

Process flow diagram or Use-case diagram



Wireframes/Mock of the proposed solution

1. Upload Page

Goal: Allow users to upload a legal document (PDF, DOCX, text) and select an output language.

A clear, centered title: "Simplify Your Legal Documents".

A large, interactive "drag and drop" zone with a prompt to "Browse Files" as an alternative.

A dropdown menu for selecting the output language (e.g., English, Hindi).

A button labeled "Simplify Document".

A disclaimer at the bottom of the page stating: "This tool is not legal advice. Always consult with a qualified lawyer."

2. Results Page

Goal: Display the simplified information from the LLM, including a summary and clause-by-clause explanations.

A prominent section at the top displaying the original document's filename.

A "Download Simplified Document" button that offers both PDF and text formats.

A heading for "**Plain-Language Summary**" followed by a bulleted list of key points.

A heading for "**Clause-by-Clause Explanation**" followed by a list of interactive cards. Each card represents a single clause.

3. Expanded Clause Card

Goal: Provide detailed, interactive information for a specific clause.

The title of the clause.

A **Simple Explanation** of the clause, typically 1-3 sentences.

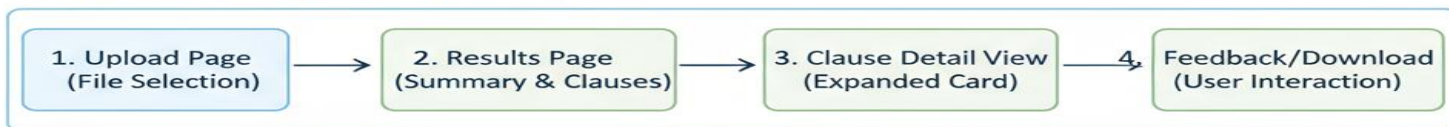
A **Risk Level** tag (e.g., High, Medium, Low), visually indicated with color.

A **Practical Example** that shows the real-world implications of the clause.

A feedback mechanism with "Correct" and "Incorrect" buttons to gather user input for future improvements.

Architecture diagram of the proposed solution

User Interface (Frontend - React)



HTTP/REST API Calls



Backend (Node.js/Express)



Data Storage



Data Layer



•Technologies to be used in the solution:

- Frontend:** React.js, Tailwind CSS (for responsive and modern UI)
- Backend:** Node.js with Express.js (for API and business logic)
- Database:** MongoDB (for storing user sessions, feedback, and saved documents)
- AI/ML Integration:**
 - OpenAI GPT / HuggingFace LLMs (for legal text simplification & Q&A)
 - NLP libraries for text chunking and clause detection
- File Processing:** pdf-parse (PDF extraction), mammoth (DOCX parsing)
- Deployment & Hosting:**
 - Vercel/Netlify (Frontend)
 - Railway/Render/Heroku (Backend)
 - MongoDB Atlas (Cloud DB)
- Version Control & Collaboration:** GitHub / GitLab
- Optional Enhancements:**
 - LangChain (for document chunking & contextual queries)
 - Translation APIs (for multilingual output)

Estimated implementation cost (optional):

- **AI/LLM API Usage (OpenAI / HuggingFace):**

~₹3,000 – ₹5,000/month (for 500–1,000 document simplifications).

- **Cloud Hosting (Backend + DB):**

~₹1,000 – ₹2,000/month (Railway/Render + MongoDB Atlas free tier can reduce cost).

- **Frontend Hosting (Vercel/Netlify):**

Free tier available; ~₹500/month if scaled.

- **Development & Tools:**

Mostly open-source (React, Node, Express, pdf-parse, mammoth, TailwindCSS).

- ☐ **Total Prototype Cost:** ~₹5,000 – ₹7,500/month (≈ \$60–90)

(Can be reduced further with free-tier cloud services and limited API usage.)

Add as per the requirements for the hackathon:

Innovation & Relevance:

- Addresses a real-world problem of legal complexity that affects citizens, students, and small businesses.
- Uses Generative AI in a novel way: **legal-context simplification with risk analysis**.

•Social Impact:

- Promotes **legal awareness and inclusivity** by offering multilingual explanations (English, Hindi, Marathi).
- Empowers users to make **informed decisions** without always relying on expensive legal consultations.
- Reduces risk of fraud and misunderstandings in contracts, policies, and government schemes.

•Scalability:

- Can be extended to multiple languages, voice support, and integration with government portals or legal helpdesks.
- Future versions can integrate **lawyer marketplaces** for expert validation alongside AI explanations.

•Hackathon Fit:

- Demonstrates **practical application of GenAI** to a critical sector.
- Achievable as a working **prototype within hackathon timeframe** using MERN + LLM APIs.
- Balances **technical feasibility and social impact**, aligning with hackathon judging criteria.

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Thank you